

PHK Early Warning Dashboard

Technical Notes

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1 Methodology for Provincial PHK Projection

1.1 Objective

The objective of the model is to project the monthly number of layoffs (PHK) at the provincial level as an early warning indicator of labour market deterioration. The model combines structural characteristics of provincial labour markets with high-frequency macroeconomic indicators to capture both long-term labour market vulnerability and short-term economic shocks.

1.2 Econometric Approach

- The dependent variable is the monthly number of reported layoffs (PHK) by province, which is a non-negative count variable with a substantial number of zero observations.
- A two-way Fixed Effects (FE) Poisson regression is employed:

$$E(PHK_{it}|X_{it}) = \exp(\beta X_{it} + \alpha_i + \gamma_t),$$

where:

- PHK_{it} denotes the number of layoffs in province i during month t ;
 - α_i represents province fixed effects;
 - γ_t represents month fixed effects;
 - X_{it} contains structural and macroeconomic variables.
- Robust standard errors are clustered at the provincial level to account for within-province serial correlation.

1.3 Rationale for Model Selection

- Provincial labour markets differ systematically in industrial composition, labour institutions, economic structure, and reporting practices. Since these unobserved characteristics are likely correlated with macroeconomic variables, the assumptions underlying Random Effects estimation are unlikely to hold. Fixed Effects therefore provide more consistent estimates by controlling for time-invariant provincial heterogeneity (Hausman, 1978; Wooldridge, 2010).
- Compared with time-series approaches such as Vector Autoregression (VAR), panel Fixed Effects models exploit both cross-sectional and time-series variation while accommodating heterogeneous provincial characteristics and relatively short monthly panels (Baltagi, 2021; Sims, 1980).
- Poisson regression is preferred because the dependent variable represents monthly counts of layoffs rather than continuous outcomes or proportions. Compared with OLS, Poisson guarantees non-negative predictions and naturally accommodates zero observations. Compared with Fractional Logit, which is designed for fractional outcomes bounded between zero and one, Poisson is more appropriate for modelling event counts. Poisson Fixed Effects estimators remain consistent under relatively weak assumptions on the conditional mean and are robust to many forms of

heteroskedasticity (Hausman et al., 1984; Santos Silva and Tenreyro, 2006; Cameron and Trivedi, 2013; Papke and Wooldridge, 1996).

1.4 Variables

Dependent Variable

- Monthly provincial PHK flow (number of reported layoffs).

Structural Variables* (Annual)

- Gross Regional Domestic Product (GRDP, log)
- Manufacturing share of GRDP
- Manufacturing employment share
- Formal employment share
- Provincial minimum wage (log)

Macroeconomic Trigger Variables* (Monthly)

- *Global*: Brent crude oil price.
- *National*: BI policy rate, IDR/USD exchange rate, Manufacturing PMI, Producer Price Index (IHPB).
- *Provincial*: Consumer Price Index (CPI), Non-Performing Loans (NPL), export value, and import value.

1.5 Lag Specification and Model Selection

- Macroeconomic trigger variables are evaluated using four alternative lag structures (1, 3, 6, and 12 months) to capture delayed responses of firms to economic shocks, while structural variables enter contemporaneously.
- (TBD) Candidate models are estimated using the training sample (2023–2025) and evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) on the validation sample.
- (TBD) The specification with the lowest prediction error is selected as the preferred forecasting model and subsequently re-estimated using the full estimation sample before generating provincial and national PHK projections.

2 Results

2.1 Poisson Regression - Coefficient

Table 1: Poisson regression estimates for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
formal_lab_share	-0.033 (0.102)	-0.085 (0.080)	-0.152 (0.131)	-0.176 (0.152)
manuf_emp_share	-0.256 (0.196)	-0.210 (0.208)	-0.042 (0.216)	-0.044 (0.439)
agri_emp_share	0.038 (0.089)	0.071 (0.091)	0.082 (0.137)	-0.111 (0.148)
min_wage_yoy	0.084* (0.037)	0.082 (0.053)	-0.013 (0.090)	0.125 (0.118)
pdrb_yoy	0.103 (0.190)	0.127 (0.120)	0.080 (0.158)	0.056 (0.048)
manuf_pdrb_share	-0.019 (0.124)	-0.066 (0.100)	-0.051 (0.114)	0.131 (0.243)
Macroeconomic Trigger Variables				
L_log_price_brent	0.709 (1.806)	2.118 (1.153)	-1.046 (2.179)	5.895 (3.848)
L_bi_rate	0.603 (0.504)	0.663* (0.294)	1.493** (0.567)	3.098*** (0.631)
L_log_fx_idr_usd	3.887 (5.216)	-2.132 (2.939)	-6.158 (7.009)	-33.984*** (8.013)
L_pmi_manuf	-0.057 (0.052)	-0.035 (0.063)	-0.068 (0.069)	0.001 (0.177)
L_ihpb_yoy	-0.067 (0.194)	-0.463** (0.174)	0.349** (0.130)	0.074 (0.469)
L_cpi_yoy	0.002 (0.125)	0.153 (0.113)	-0.461*** (0.120)	-0.333 (0.261)
L_npl_yoy	0.004 (0.004)	-0.001 (0.005)	0.003 (0.006)	-0.008 (0.006)
L_log_export	-0.254 (0.261)	-0.893* (0.440)	0.011 (0.381)	0.246 (0.402)
L_log_import	-0.225* (0.105)	-0.111 (0.098)	-0.025 (0.104)	0.152 (0.083)
Observations	981	924	843	672
DV Mean	229	231	238	244
Pseudo R ²	0.584	0.601	0.616	0.656
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of layoffs (PHK) by province. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)-(4) report Poisson regression estimates using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

2.2 Poisson Regression - IRR

Table 2: Incidence Rate Ratios (IRR) from Poisson regressions for provincial layoff projections

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Structure Variables				
formal_lab_share	0.97 (0.10)	0.92 (0.07)	0.86 (0.11)	0.84 (0.13)
manuf_emp_share	0.77 (0.15)	0.81 (0.17)	0.96 (0.21)	0.96 (0.42)
agri_emp_share	1.04 (0.09)	1.07 (0.10)	1.08 (0.15)	0.90 (0.13)
min_wage_yoy	1.09* (0.04)	1.09 (0.06)	0.99 (0.09)	1.13 (0.13)
pdrb_yoy	1.11 (0.21)	1.14 (0.14)	1.08 (0.17)	1.06 (0.05)
manuf_pdrb_share	0.98 (0.12)	0.94 (0.09)	0.95 (0.11)	1.14 (0.28)
Macroeconomic Trigger Variables				
L_log_price_brent	2.03 (3.67)	8.31 (9.59)	0.35 (0.77)	363.36 (1398.39)
L_bi_rate	1.83 (0.92)	1.94* (0.57)	4.45** (2.52)	22.16*** (13.99)
L_log_fx_idr_usd	48.75 (254.29)	0.12 (0.35)	0.00 (0.01)	0.00*** (0.00)
L_pmi_manuf	0.95 (0.05)	0.97 (0.06)	0.93 (0.06)	1.00 (0.18)
L_ihpb_yoy	0.93 (0.18)	0.63** (0.11)	1.42** (0.18)	1.08 (0.50)
L_cpi_yoy	1.00 (0.13)	1.17 (0.13)	0.63*** (0.08)	0.72 (0.19)
L_npl_yoy	1.00 (0.00)	1.00 (0.00)	1.00 (0.01)	0.99 (0.01)
L_log_export	0.78 (0.20)	0.41* (0.18)	1.01 (0.39)	1.28 (0.51)
L_log_import	0.80* (0.08)	0.89 (0.09)	0.98 (0.10)	1.16 (0.10)
Observations	981	924	843	672
DV Mean	229	231	238	244
Pseudo R ²	0.584	0.601	0.616	0.656
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: Reported coefficients are Incidence Rate Ratios (IRRs), obtained by exponentiating the estimated Poisson coefficients. An IRR greater than one indicates that an increase in the explanatory variable is associated with a higher expected number of layoffs, while an IRR below one indicates a lower expected number of layoffs, holding other variables constant. Robust standard errors clustered at the province level are reported in parentheses. Columns (1)–(4) report specifications using macroeconomic trigger variables lagged by 1, 3, 6, and 12 months, respectively. Annual structural variables are included contemporaneously in all specifications. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

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